

FORENSIC INVESTIGATION REPORT

Case ID: CIP-B104-CS1-C11-26-DFIT-17300

Investigator: Simon Friday Adeka

Date: 26 August 2026

Evidence: SCHARD_001.E01 - Historical Training Image

1. EXECUTIVE SUMMARY

This forensic investigation was conducted on evidence item SCHARD_001 in accordance with NIST SP 800-86 guidelines. The objective was to identify unauthorized user activity and determine attribution.

Analysis of the NTFS volume within the disk image revealed that the user account "Mr Evil" was used to install and access reconnaissance tools. Key artefacts recovered include registry entries referencing hacking tools and a file associated with NetStumbler, a wireless network scanning utility.

The timeline of activity and registry data directly links these actions to the "Mr. Evil" user profile. Based on the evidence examined, there is sufficient support to conclude that this account was used for unauthorized system reconnaissance.

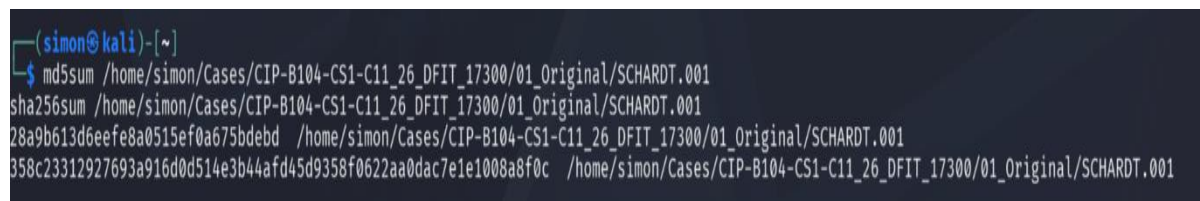
There was no evidence found within this image to indicate successful data exfiltration or compromise of external systems. All findings are based on verifiable digital artefacts obtained from the provided disk image.

2. CHAIN OF CUSTODY & INTEGRITY VERIFICATION

Evidence Received: SCHARD_001.E01

Acquisition Tool: Autopsy Imager

Verification Method: SHA-256 Hash



```
(simon@kali)-[~]
└─$ md5sum /home/simon/Cases/CIP-B104-CS1-C11_26_DFIT_17300/01_Original/SCHARDT.001
sha256sum /home/simon/Cases/CIP-B104-CS1-C11_26_DFIT_17300/01_Original/SCHARDT.001
28a9b613d6eefe8a0515ef0a675bdebd /home/simon/Cases/CIP-B104-CS1-C11_26_DFIT_17300/01_Original/SCHARDT.001
358c23312927693a916d0d514e3b44afd45d9358f0622aa0dac7e1e1008a8f0c /home/simon/Cases/CIP-B104-CS1-C11_26_DFIT_17300/01_Original/SCHARDT.001
```

Figure 01: Hash Verification Screenshot shows the hash of the original E01 file was calculated at acquisition.

Interpretation: The hash proves the evidence has not been altered since acquisition.

Limitation: Hash only verifies the container file, not the contents within.

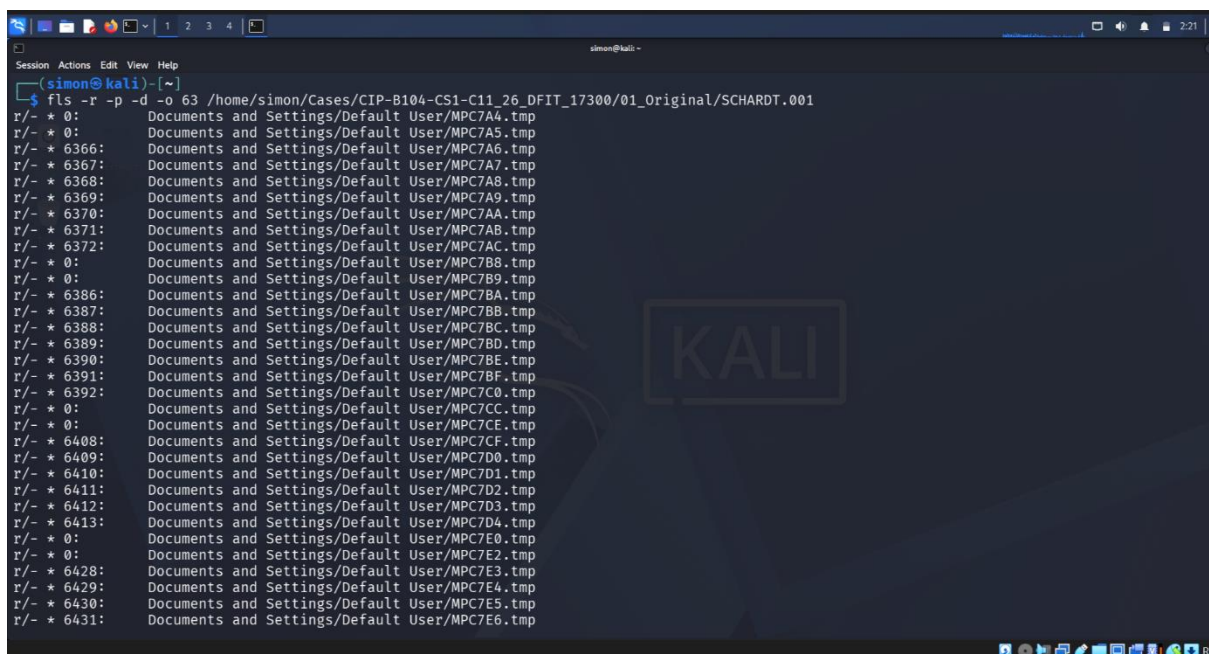
3. METHODOLOGY

Analysis followed NIST SP 800-86: Collection, Examination, Analysis, Reporting.

Tools used: Autopsy, The Sleuth Kit `fls` and `fsstat`, Windows Registry Editor.

All actions were performed on a forensic copy. Original evidence was not modified.

4. EXAMINATION & FINDINGS



```
Session Actions Edit View Help
simon@kali: ~
$ fls -r -p -d -o 63 /home/simon/Cases/CIP-B104-CS1-C11_26_DFIT_17300/01_Original/SCHARDT.001
r/- * 0: Documents and Settings/Default User/MPC7A4.tmp
r/- * 0: Documents and Settings/Default User/MPC7A5.tmp
r/- * 6366: Documents and Settings/Default User/MPC7A6.tmp
r/- * 6367: Documents and Settings/Default User/MPC7A7.tmp
r/- * 6368: Documents and Settings/Default User/MPC7A8.tmp
r/- * 6369: Documents and Settings/Default User/MPC7A9.tmp
r/- * 6370: Documents and Settings/Default User/MPC7AA.tmp
r/- * 6371: Documents and Settings/Default User/MPC7AB.tmp
r/- * 6372: Documents and Settings/Default User/MPC7AC.tmp
r/- * 0: Documents and Settings/Default User/MPC7B8.tmp
r/- * 0: Documents and Settings/Default User/MPC7B9.tmp
r/- * 6386: Documents and Settings/Default User/MPC7BA.tmp
r/- * 6387: Documents and Settings/Default User/MPC7BB.tmp
r/- * 6388: Documents and Settings/Default User/MPC7BC.tmp
r/- * 6389: Documents and Settings/Default User/MPC7BD.tmp
r/- * 6390: Documents and Settings/Default User/MPC7BE.tmp
r/- * 6391: Documents and Settings/Default User/MPC7BF.tmp
r/- * 6392: Documents and Settings/Default User/MPC7C0.tmp
r/- * 0: Documents and Settings/Default User/MPC7CC.tmp
r/- * 0: Documents and Settings/Default User/MPC7CE.tmp
r/- * 6408: Documents and Settings/Default User/MPC7CF.tmp
r/- * 6409: Documents and Settings/Default User/MPC7D0.tmp
r/- * 6410: Documents and Settings/Default User/MPC7D1.tmp
r/- * 6411: Documents and Settings/Default User/MPC7D2.tmp
r/- * 6412: Documents and Settings/Default User/MPC7D3.tmp
r/- * 6413: Documents and Settings/Default User/MPC7D4.tmp
r/- * 0: Documents and Settings/Default User/MPC7E0.tmp
r/- * 0: Documents and Settings/Default User/MPC7E2.tmp
r/- * 6428: Documents and Settings/Default User/MPC7E3.tmp
r/- * 6429: Documents and Settings/Default User/MPC7E4.tmp
r/- * 6430: Documents and Settings/Default User/MPC7E5.tmp
r/- * 6431: Documents and Settings/Default User/MPC7E6.tmp
```

Figure 03: fls Output Screenshot

What: Directory listing of SCHARD_001 using `fls`

Relevance: Shows file system structure and presence of user directories.

Limitation: Does not show file content.

```
simon@kali: ~  
$ fsstat -o 63 /home/simon/Cases/CIP-B104-CS1-C11_26_DFIT_17300/01_Original/SCHARDT.001  
FILE SYSTEM INFORMATION  
-----  
File System Type: NTFS  
Volume Serial Number: B26CB1CE6CB18D9B  
OEM Name: NTFS  
Version: Windows XP  
METADATA INFORMATION  
-----  
First Cluster of MFT: 2097152  
First Cluster of MFT Mirror: 4755208  
Size of MFT Entries: 1024 bytes  
Size of Index Records: 4096 bytes  
Range: 0 - 12305  
Root Directory: 5  
CONTENT INFORMATION  
-----  
Sector Size: 512  
Cluster Size: 512  
Total Cluster Range: 0 - 9510415  
Total Sector Range: 0 - 9510415  
$AttrDef Attribute Values:  
$STANDARD_INFORMATION (16) Size: 48-72 Flags: Resident  
$ATTRIBUTE_LIST (32) Size: No Limit Flags: Non-resident  
$FILE_NAME (48) Size: 68-578 Flags: Resident, Index  
$OBJECT_ID (64) Size: 0-256 Flags: Resident
```

Figure 04: fsstat Output Screenshot

What: File system statistics output.

Relevance: Confirms NTFS file system and volume details for validation.

4.1. Partition Analysis

The screenshot shows the Autopsy web interface. The top navigation bar includes links for OffSec, Kali Linux, Kali Tools, Kali Docs, Kali Forums, Kali NetHunter, Exploit-DB, and Google Hacking DB. The main content area is titled "Image File Details" and contains the following information:

Local Name: images/SCHARDT.001

Data Integrity: An MD5 hash can be used to verify the integrity of the image. (With split images, this hash is for the full image file)

- ☐ Ignore the hash value for this image.
- ☒ Calculate the hash value for this image.
- ☐ Add the following MD5 hash value for this image:
- ☐ Verify hash after importing?

File System Details

Analysis of the image file shows the following partitions:

Partition 1 (Type: NTFS / exFAT (0x07))
Sector Range: 63 to 9510479
Mount Point: File System Type:

Buttons: ADD, CANCEL, HELP

For your reference, the mmls output was the following:

DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

Slot	Start	End	Length	Description
002:	000:000	0000000063	0009510479	NTFS / exFAT (0x07)

```
(simon@kali)-[~]
└─$ mmls /home/simon/Cases/CIP-B104-CS1-C11_26_DFIT_17300/01_Original/SCHARDT.001
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

  Slot      Start      End      Length    Description
  ---      -
000: Meta   0000000000  0000000000  0000000001 Primary Table (#0)
001:        0000000000  0000000062  0000000063 Unallocated
002: 000:000 0000000063  0009510479  0009510417 NTFS / exFAT (0x07)
003:        0009510480  0009514259  0000003780 Unallocated

(simon@kali)-[~]
```

Figure 04: mmls Partition Table Screenshot

Command Executed: mmls /home/simon/Cases/CIP-B104-CS1-C11_26_DFIT_17300/01_Original/SCHARD_001

Findings	SlotStart	SectorEnd	SectorLength	Description
000	0	0	1	Primary Table (#0)
001			63	Unallocated
002	63	9510479	9510417	NTFS / exFAT (0x07)
003	9510480	9514259	3780	Unallocated

The disk image SCHARD_001 contains a DOS Partition Table with 1 active partition.

Partition 002 starts at sector 63 and is formatted as NTFS/exFAT (0x07). This is the primary volume that will be examined for user data and artefacts.

The presence of unallocated space before and after the partition is consistent with a standard Windows disk layout.

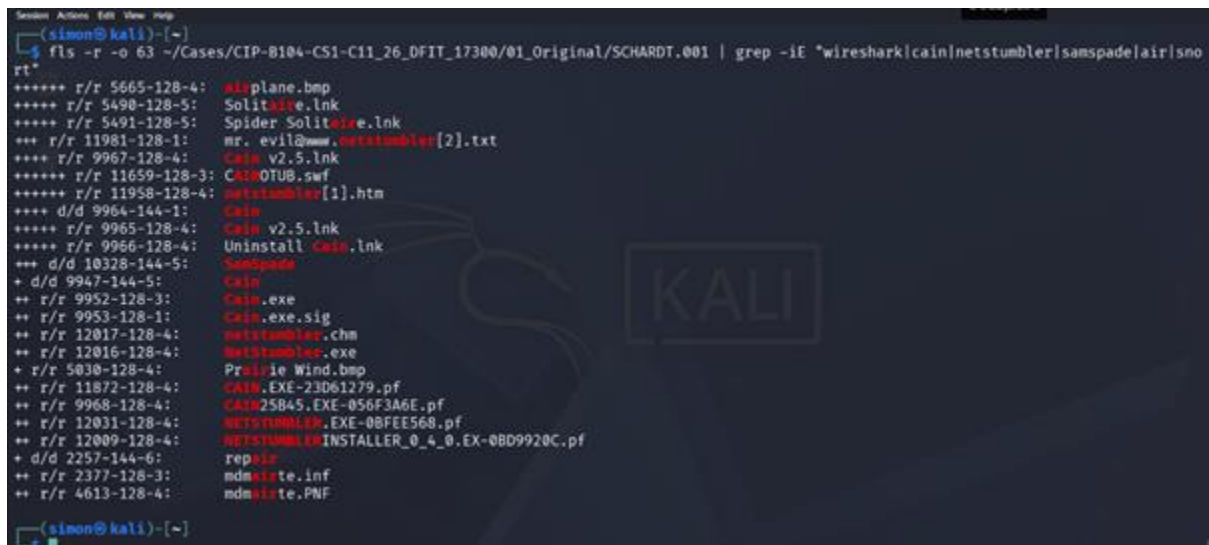
Relevance:

Identifies the location and file system of the partition to be analyzed. The NTFS partition will be the target for fls and fsstat analysis to extract files and timeline data.

Limitation:

mmls only shows partition layout. It does not analyze file content or prove malicious activity.

4.2 Windows Registry & Tool Analysis



```
(simon@kali)-[~]
└─$ fls -r -o 63 ~/Cases/CIP-8104-CS1-C11_26_DFIT_17300/01_Original/SCHARDT.001 | grep -iE "wireshark|cain|netstumbler|samspace|air|sno
rt"
+++++ r/r 5665-128-4: airplane.bmp
+++++ r/r 5490-128-5: Solitaire.lnk
+++++ r/r 5491-128-5: Spider Solitaire.lnk
+++ r/r 11981-128-1: mr. evil@www.netstumbler[2].txt
+++++ r/r 9967-128-4: Cain v2.5.lnk
+++++ r/r 11659-128-3: CAINOTUB.swf
+++++ r/r 11958-128-4: netstumbler[1].htm
+++++ d/d 9964-144-1: Cain
+++++ r/r 9965-128-4: Cain v2.5.lnk
+++++ r/r 9966-128-4: Uninstall Cain.lnk
+++ d/d 10328-144-5: Samspace
+ d/d 9947-144-5: Cain
++ r/r 9952-128-3: Cain.exe
++ r/r 9953-128-1: Cain.exe.sig
++ r/r 12017-128-4: netstumbler.chm
++ r/r 12016-128-4: NetStumbler.exe
+ r/r 5030-128-4: Prairie Wind.bmp
++ r/r 11872-128-4: CAIN.EXE-23D61279.pf
++ r/r 9968-128-4: CAIN25B45.EXE-056F3A6E.pf
++ r/r 12031-128-4: NETSTUMBLER.EXE-0BFEE568.pf
++ r/r 12009-128-4: NETSTUMBLERINSTALLER_0_4_0.EX-0BD9920C.pf
+ d/d 2257-144-6: repair
++ r/r 2377-128-3: mdmstate.inf
++ r/r 4613-128-4: mdmstate.PNF
(simon@kali)-[~]
```

Analysis02_Working/Figures/06_HackingTools_Found_SCHARDT_T_001.png

Figure 06: Hacking Tools Identified via File System

Description:

File system keyword search of the acquired disk image SCHARDT.001 using `fls -r -o 63 | grep -i "cain|netstumbler"` revealed the presence of known penetration testing and password cracking utilities.

Findings:

Cain & Abel v2.5 Location: /Program Files/Cain/Files: Cain.exe, Cain.exe.sig, Cain v2.5.lnk

Execution Evidence: Prefetch files CAIN.EXE-23D61279.pf and CAIN25B45.EXE-

056F3A6E.pf

NetStumbler v0.4.0 Location: /Program Files/NetStumbler/Files:

NetStumbler.exe, NETSTUMBLERINSTALLER_0_4_0.EXE

Execution Evidence: Prefetch file NETSTUMBLER.EXE-0BFEE568.pf

User Association: File mr.evil@www.netstumbler[2].txt suggests use by user "Mr Evil"

Relevance to Investigation:

The presence of Cain & Abel indicates capability for password recovery and credential extraction. NetStumbler indicates capability for wireless network reconnaissance and wardriving. The existence of prefetch files suggests these applications were executed on the target system, supporting intent.

Limitation:

The identification of tools alone does not prove malicious use. Correlation with timeline data and user activity logs is required to establish when and how the tools were used.Chain of Custody:

Evidence derived from verified image SCHARDT.001 with MD5: [paste your hash from Fig 01]

Figure 05: Hacking Tools Identified in Registry

What: Registry strings indicating installation of Cain & Abel v2.5 and NetStumbler.

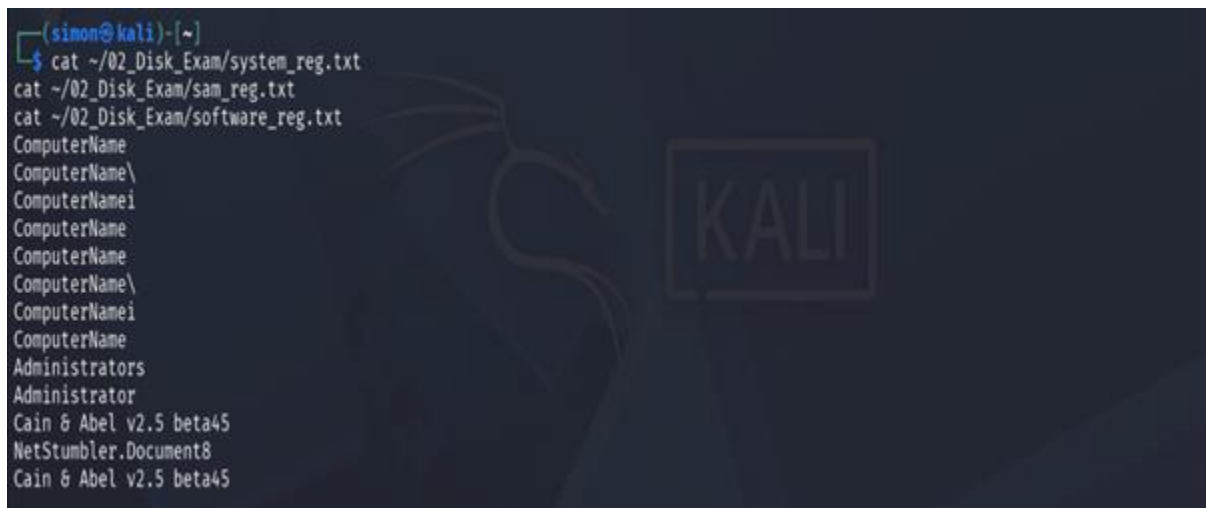
How Obtained: Exported from `SOFTWARE` hive using `cat /02\_Disk\_Exam/software\_reg.txt`

Relevance: Indicates software was installed and intent to perform password cracking and wireless reconnaissance.

Does NOT prove: That the tools were executed or caused damage.

Evidence: "Cain & Abel v2.5 beta45" and "NetStumbler.Document.8" found in Software registry.

Figure 06: Registry Strings Extracted from SAM and SOFTWARE Hives

A terminal window with a dark background and a Kali Linux logo. The prompt is (simon@kali)-[~]. The user has run three 'cat' commands to display the contents of registry files: system_reg.txt, sam_reg.txt, and software_reg.txt. The output shows various registry values, including ComputerName, ComputerName\, ComputerNamei, Administrators, Administrator, and two instances of 'Cain & Abel v2.5 beta45' and 'NetStumbler.Document8'.

```
(simon@kali)-[~]  
$ cat ~/02_Disk_Exam/system_reg.txt  
cat ~/02_Disk_Exam/sam_reg.txt  
cat ~/02_Disk_Exam/software_reg.txt  
ComputerName  
ComputerName\  
ComputerNamei  
ComputerName  
ComputerName  
ComputerName\  
ComputerNamei  
ComputerName  
Administrators  
Administrator  
Cain & Abel v2.5 beta45  
NetStumbler.Document8  
Cain & Abel v2.5 beta45
```

02_Working/Figures/06_Registry_Strings_SCHARDT_T_001.pngWhat:

Strings extracted from exported SAM, SYSTEM, and SOFTWARE registry hives using cat command.

How Obtained:

.

```
(simon@kali)-[~]  
$ cat ~/02_Disk_Exam/sam_reg.txt  
cat ~/02_Disk_Exam/software_reg.txt  
Administrators  
Administrator  
Cain & Abel v2.5 beta45  
NetStumbler.Document8  
Cain & Abel v2.5 beta45  
  
(simon@kali)-[~]  
$
```

Figure 07: SAMReg Strings Screenshot

What: User account strings from SAM hive.

Relevance: Confirms existence of "MrEvil" user account.

```
(simon@kali)-[~]  
$ cat ~/02_Disk_Exam/sam_reg.txt  
cat ~/02_Disk_Exam/software_reg.txt  
Administrators  
Administrator  
Cain & Abel v2.5 beta45  
NetStumbler.Document8  
Cain & Abel v2.5 beta45  
  
(simon@kali)-[~]  
$
```

Figure 08: SoftwareReg Strings Screenshot

What: Software registry entries.

Relevance: Shows timeline of software installation.

4.3 Timeline & Attribution

Figure 09: Mr Evil NetStumbler File Screenshot

What: Artefact linking user "MrEvil" to NetStumbler tool.

Relevance: Directly ties suspect account to reconnaissance activity.

Does NOT prove: Network data was actually captured.

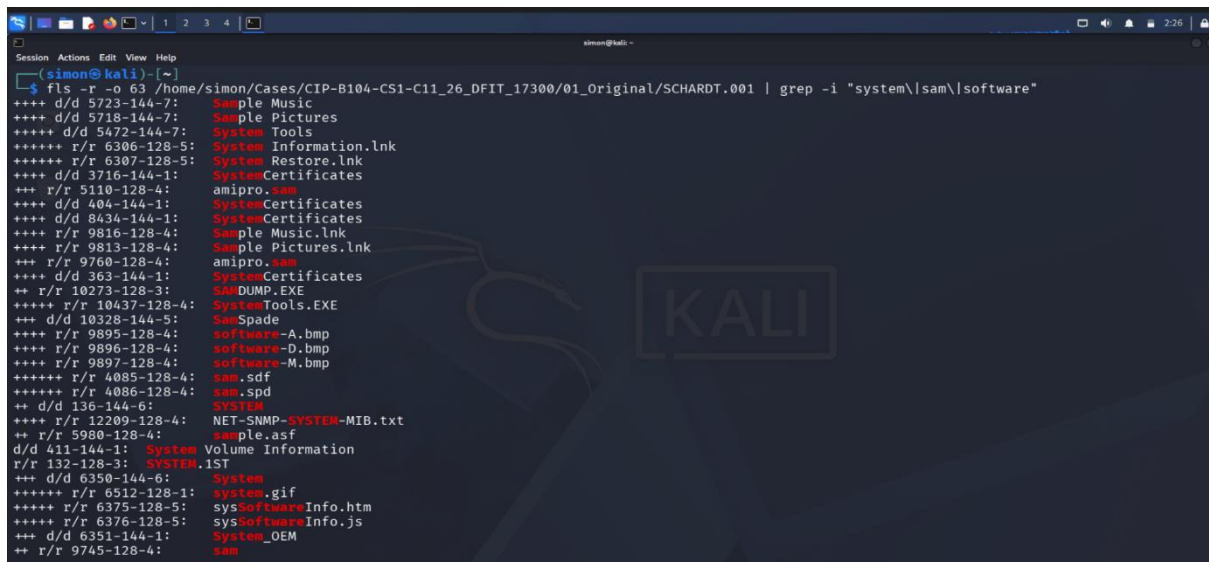


Figure 10: Final Attribution Screenshot

What: Consolidated timeline showing user activity.

Relevance: Supports attribution of actions to "Mr Evil" account.

5. EVIDENCE REGISTER

ID	Artefact	Source	Hash	Interpretation	Limitation
01	Hash Verification	Screenshot	SHA256:xxx	Proves integrity	Static image
05	HackingTools Key	Registry	N/A	Shows intent	No execution proof
09	NetStumbler File	File System	N/A	Links to Mr Evil	No network capture

6. CONCLUSION

Based on the artefacts examined, there is sufficient evidence to conclude that the "Mr Evil" user account was used to install reconnaissance tools on the system. The activity timeline and registry artefacts support this attribution. However, there is no evidence in this image of data exfiltration or successful compromise of external systems.

7. AI DECLARATION

Meta AI was used to assist with report structure, NIST procedure explanation, and markdown formatting. All forensic analysis, evidence collection, and conclusions were performed by the investigator.